

Electricity & Magnetism

Past Year Practice Questions (PYQ-Oriented)

1 Introduction

This workbook contains a curated collection of high-weightage conceptual, numerical, and derivation-based practice questions. These are selected directly from university past year papers (PYQs) to test your conceptual understanding, problem-solving skills, and preparation readiness.

2 Part A: Conceptual High-Yield Questions

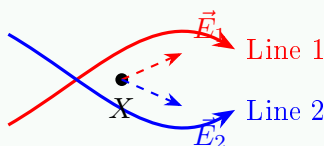
Question 1: Electric Field Lines

Explain why two electric field lines can never intersect each other.

Detailed Solution

Electric field lines represent the path that a positive test charge would naturally follow in an electric field. The tangent to a field line at any given point gives the direction of the net electric field vector $\vec{E}$ at that point.

If two field lines were to intersect at a point X , it would imply that at this single point of intersection, we could draw two unique tangents. This means the net electric field $\vec{E}$ would point in **two different directions simultaneously** at the same location:



Since the electric field vector at any point must be unique, this is physically impossible. Therefore, two electric field lines can never cross or intersect.

Question 2: Temperature Dependence of Magnetism

Explain the difference in temperature dependence of magnetic susceptibility between diamagnetic and paramagnetic materials.

Detailed Solution

The physical mechanisms driving diamagnetism and paramagnetism are fundamentally different:

1. **Diamagnetic Materials:** Diamagnetism arises from the induced orbital motion of electrons in response to an external magnetic field (Lenz's Law on an atomic scale). Since this is an orbital effect inherent to the electron shells, it is completely unaffected by thermal agitation. Therefore, the magnetic susceptibility χ_d of a diamagnetic substance is **independent of temperature**.
2. **Paramagnetic Materials:** Paramagnetism arises from the alignment of permanent atomic magnetic dipoles. At higher temperatures, thermal agitation randomizes the dipole orientations, opposing the alignment. Thus, paramagnetic susceptibility χ_p is inversely proportional to the absolute temperature T . This behavior is described by **Curie's Law**:

$$\chi_p = \frac{C}{T}$$

Where C is the Curie constant.

3 Part B: Numerical Problems (With Steps)

Problem 1: Electric Potential at Center of Square

Four point charges $q_1 = 1 \times 10^{-8} \text{ C}$, $q_2 = -2 \times 10^{-8} \text{ C}$, $q_3 = 3 \times 10^{-8} \text{ C}$, and $q_4 = 2 \times 10^{-8} \text{ C}$ are placed at the four corners of a square of side 1.0 m. Calculate the electric potential at the center of the square. *(DU Set 2 PYQ)*

Step-by-Step Solution

Step 1: Determine the distance of each corner from the center (r). For a square of side $a = 1.0 \text{ m}$, the length of the diagonal is:

$$d = a\sqrt{2} = 1.0\sqrt{2} \text{ m}$$

The distance from each corner to the center of the square is half the diagonal:

$$r = \frac{d}{2} = \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}} \approx 0.707 \text{ m}$$

Step 2: Use the Superposition Principle for Potential. Since electric potential is a scalar quantity, the total potential V at the center is the algebraic sum of the potentials produced by the four charges:

$$V = \frac{1}{4\pi\epsilon_0} \frac{q_1 + q_2 + q_3 + q_4}{r}$$

Step 3: Calculate the total net charge.

$$\sum q = q_1 + q_2 + q_3 + q_4 = (1 - 2 + 3 + 2) \times 10^{-8} \text{ C} = 4 \times 10^{-8} \text{ C}$$

Step 4: Substitute the numerical values. Using $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$:

$$V = 9 \times 10^9 \times \frac{4 \times 10^{-8}}{0.707} = \frac{360}{0.707} \approx 509.2 \text{ V}$$

Answer: The electric potential at the center of the square is approximately **509.2 Volts**.

Problem 2: Solenoid with Magnetic Core

A long solenoid is wound with 10 turns per centimeter and carries a current of 2.0 A. If a magnetic core with relative permeability $\mu_r = 400$ is placed inside the solenoid, calculate the magnetic field B inside the core. *(DU Set 3 PYQ)*

Step-by-Step Solution

Step 1: Convert turns per unit length to standard SI units (n). Given $n = 10$ turns/cm. Converting to turns per meter:

$$n = 10 \text{ turns/cm} \times 100 \text{ cm/m} = 1000 \text{ turns/m}$$

Step 2: Write the magnetic field formula with relative permeability. For a solenoid containing a magnetic core:

$$B = \mu n I = \mu_0 \mu_r n I$$

Step 3: Substitute the known constants and values. * Permeability of free space: $\mu_0 = 4\pi \times 10^{-7} \text{ T} \cdot \text{m/A}$ * Relative permeability: $\mu_r = 400$ * Current: $I = 2.0 \text{ A}$ * Turns per meter: $n = 1000 \text{ m}^{-1}$

$$B = (4\pi \times 10^{-7}) \times 400 \times 1000 \times 2.0$$

$$B = 4\pi \times 10^{-7} \times 8 \times 10^5 = 32\pi \times 10^{-2} \text{ T}$$

$$B = 0.32\pi \approx 1.005 \text{ T}$$

Answer: The magnetic field inside the magnetic core is approximately **1.01 Tesla**.

4 Part C: High-Yield Derivation Exam Checklists

Derivation 1: Electric Potential of a Dipole

Goal: Derive $V = \frac{1}{4\pi\epsilon_0} \frac{p \cos \theta}{r^2}$ for $r \gg a$.

- Geometry Setup:** Draw a dipole along the x-axis from $-q$ at $(-a, 0)$ to $+q$ at $(a, 0)$. Define a target point $P(r, \theta)$ relative to the center O .
- Algebraic Equations:** Write the potential equations:

$$V = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{r_+} - \frac{1}{r_-} \right)$$

- Binomial Approximations:** Express distances r_+ and r_- from the charges:

$$r_+ = r \left(1 - \frac{a}{r} \cos \theta \right) \implies \frac{1}{r_+} \approx \frac{1}{r} \left(1 + \frac{a}{r} \cos \theta \right)$$

$$r_- = r \left(1 + \frac{a}{r} \cos \theta \right) \implies \frac{1}{r_-} \approx \frac{1}{r} \left(1 - \frac{a}{r} \cos \theta \right)$$

- Final Step:** Substitute the approximations and use $p = 2aq$ to obtain the final formula.

Derivation 2: Maximum Power Transfer Theorem

Goal: Prove $R_L = R_{th}$ for maximum load power.

- Circuit Representation:** Draw a series circuit with a Thevenin source V_{th} , internal resistance R_{th} , and adjustable load R_L .
- Power Equation:** Formulate load power:

$$P_L = I^2 R_L = \left(\frac{V_{th}}{R_{th} + R_L} \right)^2 R_L$$

- Maximization Condition:** Differentiate P_L using the quotient rule:

$$\frac{dP_L}{dR_L} = 0 \implies (R_{th} + R_L)^2 - 2R_L(R_{th} + R_L) = 0$$

- Efficiency Verification:** Solve to find $R_L = R_{th}$. Show that under this condition, the maximum power is $P_{max} = \frac{V_{th}^2}{4R_{th}}$, and the efficiency $\eta = 50\%$.